

Lecture 1 : Introduction and Basic Calculus

Piotr Zwiernik

19BE02: Calculus and Linear Algebra



Course structure

Instructor: Piotr Zwiernik (piotr.zwiernik@bse.eu).

Five lectures: today, Fri, Mon, Tue, Wed.

Each day: 4 hours of lectures, 1 hour of tutorials.

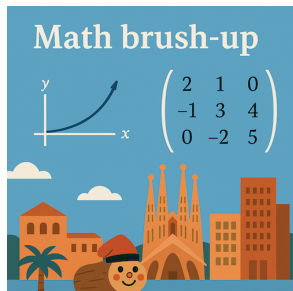
Lectures: 11-1pm and 2-4pm.

Tutorials: 4-5pm.

TA: Francisco Calvillo (francisco.calvillo@bse.eu).

Grading: This course is not graded. But there will be an exam on Friday, Sept 19th, 10-11am.

Course website: <https://pzwiernik.github.io/math-brushup/>.



Goal of the course

Review core tools of calculus that power modern economics, finance, and data-driven policy.

These tools are **crucial** in courses on micro/macro, econometrics, finance, and machine learning.

Warm up before the Master starts: sharpen intuition, not just technique.

Main references:

1. *Mathematics for Economists* by C. P. Simon and L. Blume
 2. *Mathematics of Economics and Business* by F. Werner and Y. N. Sotskov
- (any other standard textbook should be fine)

Course content

1. Sequences and series
2. Functions of one variable
3. Differentiation
4. Integration
5. Vectors
6. Matrices and determinants
7. Systems of linear equations
8. Eigenvalue problems and quadratic forms
9. Functions of several variables

Cultivate a Critical Mindset

Technical knowledge requires effort.

- **If a step feels “magical”, stop.** Ask *why* it works, not just *how*.
- **Challenge every answer — especially given by LLMs.**
Large-language models can sound confident yet be wrong; verify with first principles or a textbook.
- **Use multiple lenses.**
Check special cases, sketch a graph, test numerically, or search for a counterexample.

Welcome to Our Class Q&A Forum!

Purpose: A Space for Collaborative Learning

This forum is our dedicated space for all questions related to course content. It's a place for you to ask for help and to help your classmates.

How to Access

1. Go to the **Classwork** tab.
2. Find the topic "**Student Q&A Forum**".
3. Click on the question thread to view discussions or ask your own!

Be Active!

- **Ask:** Don't hesitate to post a question.
- **Answer:** If you know the answer to a classmate's question, please reply!

I will check the forum 2-3 times a day.

Sequences

Sequences

A **sequence** is an ordered list of real numbers $(a_1, a_2, \dots)$ indexed by $n \in \mathbb{N}$.

Notation: $\{a_n\}_{n=1}^{\infty}$ where each $a_n \in \mathbb{R}$.

Defined by a **formula** or a **recursion**.

Examples:

1. $a_n = \frac{n-1}{2n+1}$ ($a_1 = 0, a_2 = \frac{1}{5}, a_3 = \frac{2}{7}, \dots$)

2. $a_1 = 2, a_{n+1} = 2a_n - 1$ ($2, 3, 5, 9, \dots$)

3. $a_1 = 1, a_2 = 1, a_{n+1} = a_n + a_{n-1}$ (Fibonacci)

4. **Arithmetic:** $a_n = a_1 + (n-1)d$

5. **Geometric:** $a_n = a_1 r^{n-1}$

Why sequences show up in modern econ/data

Where they appear:

- **User metrics:** daily active users (DAU) over days n .
- **Inflation tracking:** monthly CPI/CPIH index a_n .
- **Training loss:** $a_n = \text{loss after } n \text{ gradient steps}$ (decreasing sequence).
- **Carbon budgets:** remaining allowance after n years.

Read Ch. 2 Werner-Sotskov. Exercises: 2.4, 2.5, 2.7 (Werner-Sotskov)

Small motivating example: Creator economy revenue

Imagine a YouTuber uploading videos one by one. Each video gets a certain number of views:

$$v_1, v_2, v_3, \dots$$

and YouTube pays the creator a fixed rate of c euros per 1,000 views (*RPM*).

The revenue from video k is

$$r_k = c \cdot \frac{v_k}{1000}.$$

The total revenue after n videos is then the sequence

$$S_n = r_1 + r_2 + \dots + r_n.$$

This is also an example of a **partial sum**; more on that in a second.

Small motivating example: EAR vs APR in FinTech

Let $a_0 = P$ be the initial investment. If a platform advertises an *annual percentage rate* (APR) R but applies interest m times per year, then after n years:

$$a_n = P \left(1 + \frac{R}{m} \right)^{mn}.$$

The one-year *effective annual rate* (EAR) is:

$$\frac{a_1 - a_0}{a_0} = \left(1 + \frac{R}{m} \right)^m - 1.$$

(APR is the quoted yearly rate ignoring compounding; APY/EAR accounts for the fact that interest itself earns interest when applied multiple times in a year.)

What is the actual daily interest? (ignoring fineprint)

Revolut

Personal

Business

Kids & Teens

Company

Log in

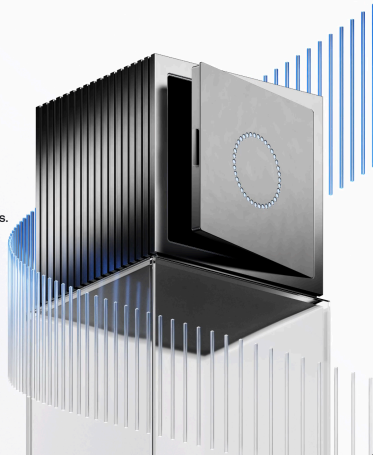
Sign up

Instant Access Savings

SAVINGS, ON DEMAND

Up to 4.5% AER (variable) interest paid daily. No withdrawal fees. No minimums. Instant access anytime.

Get started



Properties of a sequence

A sequence is **increasing** if $a_{n+1} \geq a_n$ (strictly if $>$) for all $n = 1, 2, \dots$

Decreasing if $a_{n+1} \leq a_n$ (strictly if $<$).

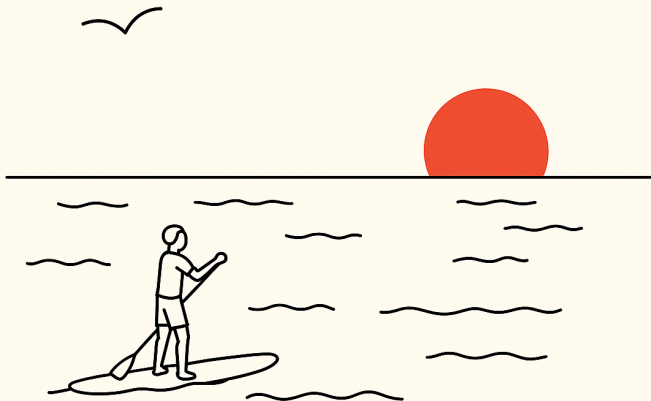
Monotone if increasing or decreasing.

Bounded if $\exists C > 0$ s.t. $|a_n| \leq C$ for all n .

Examples:

1. $a_n = 2(n-1)^2 - n$ is strictly increasing (since $a_{n+1} - a_n = 4n - 3 > 0$)
2. $a_n = 1/n^2$ is bounded ($0 < a_n \leq 1$)
3. **Training loss:** if a learning rate is sensible, $a_n = \text{loss}$ often decreases and is bounded below by 0 $\Rightarrow$ convergent in many practical cases.

Can a strictly increasing sequence be bounded?



Limit of a sequence

Limit of a sequence

A sequence $(a_n)_{n \in \mathbb{N}}$ **converges to zero** if a_n get arbitrarily close to zero.

More generally, $a \in \mathbb{R}$ is the **limit** of $(a_n)_{n \in \mathbb{N}}$ if, as $n \rightarrow \infty$, a_n gets arbitrarily close to a .

(equivalently $d_n = |a_n - a|$ converges to zero)

Notation: $\lim_{n \rightarrow \infty} a_n = a$.

A sequence is **convergent** if it has a limit; otherwise it **diverges**.

Examples

1. $a_n = \frac{1}{n^2} \rightarrow 0$
2. $a_n = (-1)^n$ has no limit
3. $a_n = n^2 \rightarrow \infty$
4. $a_n = r^n \rightarrow 0$ if $|r| < 1$; diverges if $r > 1$; no limit if $r \leq -1$.

Formal definition

How do you formalize the statement “gets arbitrarily close”?

Definition: $a = \lim_{n \rightarrow \infty} a_n$

$$\forall \epsilon > 0 \quad \exists N_\epsilon \quad \text{s.t.} \quad \forall n > N_\epsilon : \quad |a_n - a| < \epsilon.$$

Example: $a_n = \frac{1}{n} \rightarrow 0$. For any $\epsilon > 0$, take $N_\epsilon > \frac{1}{\epsilon}$.

Checks:

- $\lim a_n = a$ if and only if $\lim |a_n - a| = 0$.
- If $a_n = a$ for all n , then $\lim a_n = a$.

Theorem(Squeeze theorem):

If $a_n \leq b_n \leq c_n$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$ then $\lim_{n \rightarrow \infty} b_n = L$.

e.g. $0 < \frac{1}{2n^2-1} \leq \frac{1}{n}$ for $n \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} \frac{1}{2n^2-1} = 0$.

Squeeze theorem



Limits that matter in practice

Classics

1. $\left(1 + \frac{1}{n}\right)^n \rightarrow e$

2. $\left(1 + \frac{x}{n}\right)^n \rightarrow e^x$ for any real x

Continuous compounding & EAR: If APR is R , then

$$\lim_{m \rightarrow \infty} \left(1 + \frac{R}{m}\right)^m = e^R, \quad \text{EAR} = e^R - 1.$$

Example: APR = 8% $\Rightarrow$ EAR $\approx$ 8.329%.

With monthly compounding ($m = 12$) EAR is 8.3%.

With daily compounding ($m = 365$) EAR is 8.328%

Monotone convergence and algebra of limits

Theorem

Every bounded monotone sequence is convergent.

(Just bounded or just monotone is not enough in general)

Example: the sequence $a_1 = 1$, $a_{n+1} = \sqrt{3a_n}$ is bounded by 3 (proof by induction) and strictly increasing (since $\frac{a_{n+1}}{a_n} > 1$), thus convergent.

Algebra of limits

If $\lim a_n = a$ and $\lim b_n = b$, then

$$\lim(a_n + b_n) = a + b, \quad \lim(a_n b_n) = ab. \quad \text{Moreover, } \lim \frac{a_n}{b_n} = \frac{a}{b} \text{ if } b_n, b \neq 0.$$

Series

Partial sums and series

The n th partial sum s_n of $\{a_n\}$ is

$$s_n = \sum_{k=1}^n a_k.$$

Examples

1. Arithmetic sequence $a_n = a_1 + (n-1)d$: $s_n = na_1 + \frac{n(n-1)}{2}d$
2. Geometric sequence $a_n = a_1 r^{n-1}$, $r \neq 1$: $s_n = a_1 \frac{1-r^n}{1-r}$

Use-cases:

- Let *margin* be the profit earned per customer per period and p be the customer retention rate. The total profit from a single customer over n periods is:
 $s_n = \sum_{k=1}^n \text{margin} \cdot p^k$.
- Let E_k be the emissions produced in period k (e.g., in tonnes of CO₂). The total emissions over n periods are: $s_n = \sum_{k=1}^n E_k$, which can be compared to a fixed carbon budget.

Series and convergence

Partial sums also form a sequence. A **series** is the limit (if it exists) of partial sums:

$$\sum_{k=1}^{\infty} a_k := \lim_{n \rightarrow \infty} s_n.$$

Geometric series: converges if and only if $|r| < 1$ with

$$\sum_{k=1}^{\infty} a_1 r^{k-1} = \frac{a_1}{1-r} \quad (\text{we can show this formally}).$$

Customer Lifetime Values: Consider the example on the previous slide. With margin m and monthly retention $p < 1$, discounted at monthly rate d ,

$$\text{CLV} = \sum_{k=1}^{\infty} \frac{m p^k}{(1+d)^k} = \frac{m \frac{p}{1+d}}{1 - \frac{p}{1+d}} = \frac{mp}{1+d-p}.$$

Chapter 2: Functions of one variable

A **one-variable function** assigns a unique $y \in \mathbb{R}$ to each $x \in \mathbb{R}$: $y = f(x)$.

x : independent (exogenous) variable; y : dependent (endogenous) variable.

Examples in econ/data: cost, revenue, demand, utility, probability of purchase vs. price, click-through rate vs. ad spend, etc.

Read Ch. 3 Werner-Sotskov; Simon-Blume 2.1-2.2, 5.1-5.3.

Graph and domain

Definition

The **graph** of f is $\{(x, y) \in \mathbb{R}^2 : y = f(x)\}$.

The **domain** D is the set of x where $f(x)$ is defined; $D \subseteq \mathbb{R}$.

Notation: $f : D \rightarrow \mathbb{R}$.

Example: $f(x) = \frac{1}{x}$ has $D = \mathbb{R} \setminus \{0\}$; $f(x) = \sin(x)$, $D = \mathbb{R}$.

Sometimes we restrict the domain (e.g., prices $x \geq 0$).

Properties of functions

A function $f : D \rightarrow \mathbb{R}$ is **increasing** if $f(x_1) \leq f(x_2)$ for any $x_1 < x_2$ in D (strict if $f(x_1) < f(x_2)$).

Decreasing: reverse inequality.

Bounded: $\exists C > 0$ s.t. $|f(x)| \leq C$ for all $x \in D$.

Examples:

- **Price-demand curve** often decreasing on realistic intervals (then zero).
- **Learning curve** typically decreasing as algorithm progresses.

Convex/concave functions

For $x_1 < x_2$ and $t \in [0, 1]$, the **convex combination** is $tx_1 + (1 - t)x_2$.

A set $D \subset \mathbb{R}$ is **convex** if it contains convex combinations.

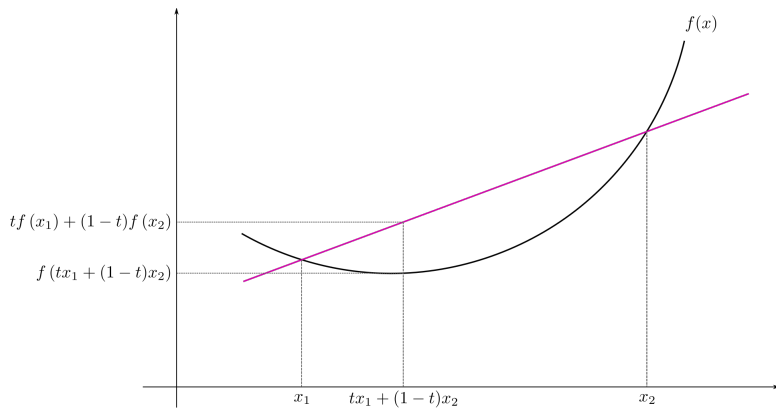
A function $f : D \rightarrow \mathbb{R}$ on convex D is **convex** if

$$f(tx_1 + (1 - t)x_2) \leq tf(x_1) + (1 - t)f(x_2)$$

(and **concave** if $\geq$).

For the strict version replace $\leq$ by $<$ for $t \in (0, 1)$ and $x_1 \neq x_2$.

Convex function



Quasi-convex/concave functions

f is quasi-convex if

$$f(tx_1 + (1 - t)x_2) \leq \max\{f(x_1), f(x_2)\}.$$

Quasi-concave if $\geq \min\{f(x_1), f(x_2)\}$.

Facts: convex $\Rightarrow$ quasi-convex; concave $\Rightarrow$ quasi-concave.

Any strictly increasing function is both quasi-convex and quasi-concave.

Economic examples:

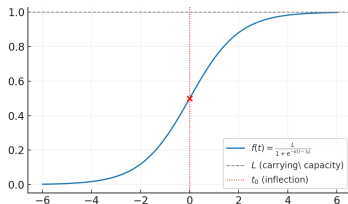
- **Indifference curves:** If a consumer is indifferent between two bundles, any convex combination is at least as good as one of them. This gives quasi-convex preferences.
- **Profits in price:** As a function of price, profits often rise, then fall (too cheap or too expensive reduces profit). This bell-shaped behavior is quasi-concave.

Economic example: logistic growth (adoption curves)

The logistic function with $L > 0$, $k > 0$, $t_0 \in \mathbb{R}$:

$$f(t) = \frac{L}{1 + e^{-k(t-t_0)}}$$

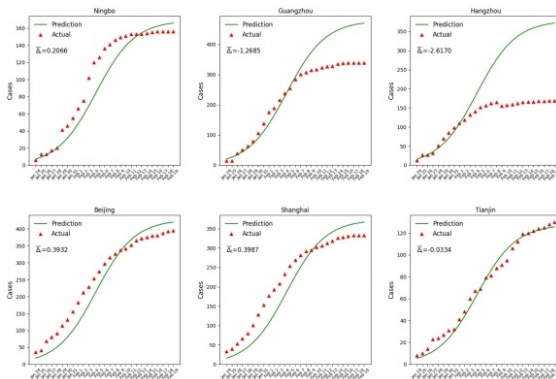
models **technology/product adoption** or **viral spread**. Early exponential growth, mid-phase inflection at t_0 , saturation near L (market size).

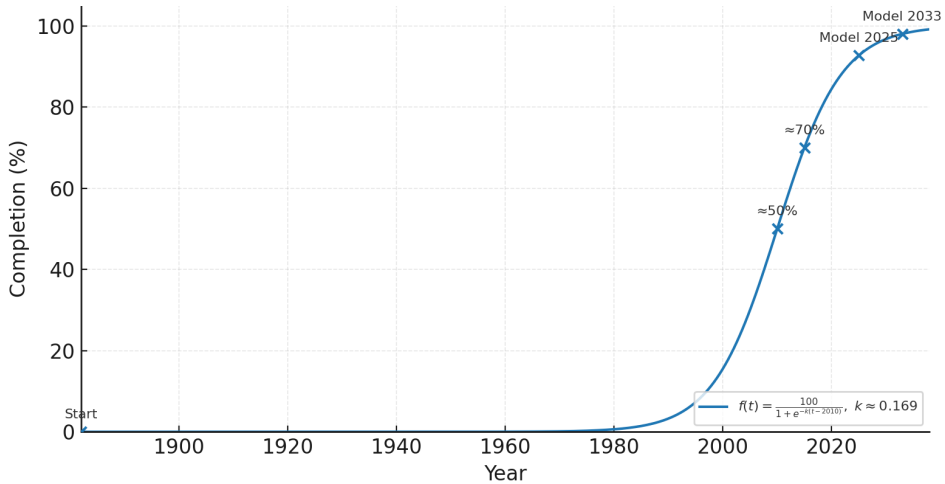


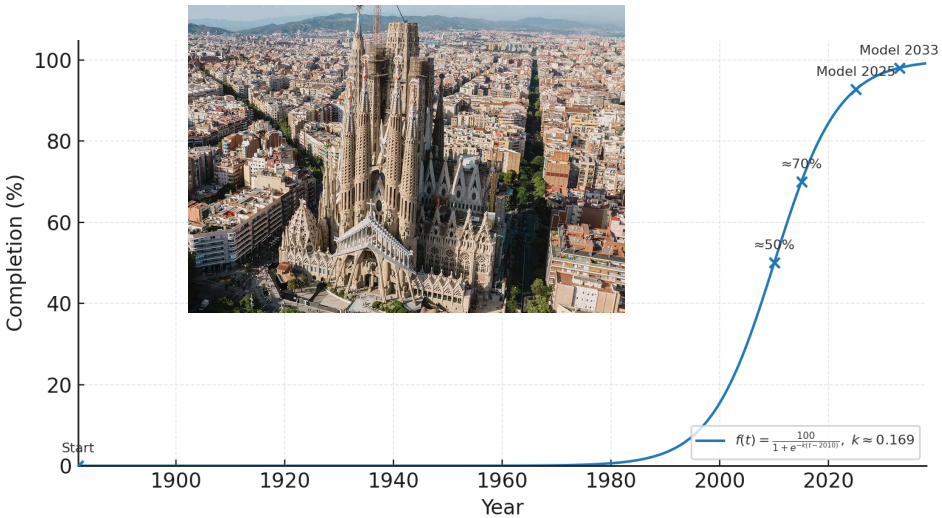
Interpretation: growth accelerates, then slows as saturation/frictions (capacity, regulation, budgets) bind. Same shape often appears in **learning curves** (accuracy vs. data).

Epidemics and beyond

Logistic curves fit epidemic growth, but also market penetration, platform adoption, content diffusion. Check the following epidemic curves over time of the COVID-19 in China







Examples of functions

1. **Monomials:** $f(x) = ax^k$, $a \in \mathbb{R}$.
2. **Polynomials:** finite sums of monomials.
3. **Rational:** ratios of polynomials.
4. **Exponential:** $f(x) = a^x$, $a > 0$.
5. **Trigonometric:** $\sin x$, $\cos x$, $\dots$
6. **Linear:** $f(x) = ax + b$ (slope a , intercept b ; also $y - f(x_0) = a(x - x_0)$).
7. **Natural exponential:** $f(x) = e^x$ ($e^x = \lim_n (1 + \frac{x}{n})^n$).
8. **Natural log:** $f(x) = \log x$ ($e^{\log x} = x$, $\log(e^x) = x$)

Economic examples (often based on the above fundamental examples):

- **Demand curves:** typically decreasing in price (law of demand).
- **Cost curves:** often convex, since producing extra units gets more expensive at higher scale.
- **Utility functions:** often concave, to capture *diminishing marginal utility* (each extra unit of a good adds less satisfaction).
- **Log-sum-exp:** arises in discrete choice models; connects probabilities to utilities.

Exponential functions

$$f(x) = 1^x$$

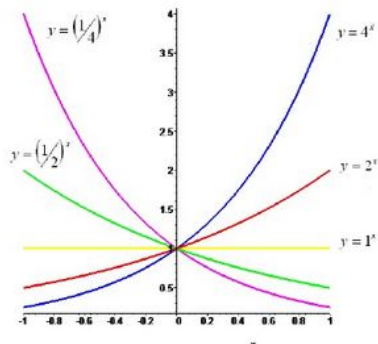
Horizontal line with y-intercept 1

$$f(x) = a^x, \quad 0 < a < 1$$

Exponential decay

$$f(x) = a^x, \quad a > 1$$

Exponential growth



Chapter 3: Differentiation

Differentiation measures sensitivity: how $f(x)$ changes with small changes in x

$$f(x + \Delta x) \approx f(x) + f'(x)\Delta x \quad \text{or, equivalently,} \quad \frac{f(x + \Delta x) - f(x)}{\Delta x} \approx f'(x).$$

Examples: price $\rightarrow$ demand sensitivity, ad spend $\rightarrow$ clicks, carbon tax $\rightarrow$ emissions, risk $\rightarrow$ portfolio value.

Calculus also underpins extrema, monotonicity, convexity — and the local behavior behind optimization and ML.

Read Ch. 4 Werner-Sotkov; Simon-Blume Chs. 2-5.

Exercises: 4.6(c), 4.10, 4.14, 4.15(a)-(c), 4.16(d), 4.17(b) (Werner-Sotkov), 5.5 (b)-(c)-(e) (Simon-Blume)

Functional Limits and Continuity

Limit of a function

Let $f : D \rightarrow \mathbb{R}$ and $x_0 \in \mathbb{R}$. We write $L = \lim_{x \rightarrow x_0} f(x)$ if

$$x_n \rightarrow x_0 \implies y_n := f(x_n) \rightarrow L.$$

Remark: x_0 need not be in D (removable/essential discontinuities matter in econ too).

Example: $\lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$ since $0 \leq |x \sin \frac{1}{x}| \leq |x|$.

Algebra of limits

If $\lim_{x \rightarrow x_0} f(x) = y_1$ and $\lim_{x \rightarrow x_0} g(x) = y_2$, then

$$\lim_{x \rightarrow x_0} (f(x) + g(x)) = y_1 + y_2, \quad \lim_{x \rightarrow x_0} (f(x)g(x)) = y_1 y_2, \quad \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{y_1}{y_2} \text{ if } y_2 \neq 0.$$

Continuity

Definition

f is **continuous** at $x_0 \in D$ if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. If f is continuous at every point of $B \subset D$, we say f is continuous on B .

Theorem: If f, g are continuous at x_0 then so are $f + g$ and fg . If moreover, $g(x) \neq 0$ around x_0 , then $\frac{f}{g}$ is continuous at x_0 .

Heine-Borel fact (1D): continuous $f : [a, b] \rightarrow \mathbb{R}$ is bounded.

Non-example

Consider the function $f(x) = \frac{|x|}{x}$ for $x \neq 0$ and $f(0) = 1$. This function is not continuous. Why?

Composition and inverse

Two important results

(composition) If f, g are continuous, so is $h(x) = g(f(x))$ (we also write $h = g \circ f$); moreover

$$\lim_{x \rightarrow x_0} g(f(x)) = g\left(\lim_{x \rightarrow x_0} f(x)\right).$$

(inverse) If $f : [a, b] \rightarrow \mathbb{R}$ is strictly monotone and continuous with $f(a) = c, f(b) = d$, then the inverse $f^{-1} : [c, d] \rightarrow [a, b]$ is also strictly monotone and continuous.

Economic example: p = price of a good, q = quantity sold. The *demand function* $q(p)$ gives the quantity q consumers will buy at price p . Equivalently, the *inverse demand function* $p(q)$ gives the price at which q units can be sold. Revenue as a function of quantity is:

$$R(q) = p(q) \cdot q.$$



Differentiation

Derivative

Let $f : (a, b) \rightarrow \mathbb{R}$. f is differentiable at x if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

exists. Denote this limit by $f'(x)$.

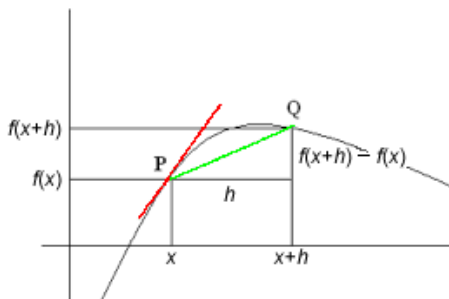
Fact: differentiable $\Rightarrow$ continuous.

The converse does not hold: $|x|$ is continuous but not differentiable at 0 (why?).

Geometric interpretation

If f is differentiable at x_0 , the tangent line to the graph of f at $(x_0, f(x_0))$ is

$$y = f(x_0) + f'(x_0)(x - x_0).$$



slope of **secant** through P, Q : $\frac{f(x+h)-f(x)}{h}$; slope of **tangent** at P : $f'(x)$

A “marginal” is a derivative

In economics, a **marginal** quantity measures how much something changes when we increase an input by a small amount. Mathematically, we often approximate

$$f(x + 1) - f(x) \approx f'(x),$$

when the change is 1 unit.

Examples:

- **Marginal revenue:** $\frac{d}{dq}[p(q) q]$ at the current sales level q . The derivative tells you how much extra revenue you gain (or lose) by selling one more unit.
- **Click-through sensitivity:** $\frac{d \text{CTR}}{d(\text{ad spend})}$ at the budget currently in use. Here CTR (click-through rate) is a function of ad spending; the derivative measures how much CTR improves with an extra euro of spending.

If the change is Δx rather than 1: $f(x + \Delta x) - f(x) \approx f'(x) \Delta x$.

In differential notation: $df \approx f'(x) dx$

Derivatives of elementary functions

1. $c' = 0$
2. $(x^n)' = nx^{n-1}$ for $n \in \mathbb{N}$
3. $(x^\alpha)' = \alpha x^{\alpha-1}$ for $\alpha \in \mathbb{R}, x > 0$
4. $(\log x)' = \frac{1}{x}, x > 0$
5. $(\sin x)' = \cos x$
6. $(\cos x)' = -\sin x$
7. $(e^x)' = e^x$
8. $(a^x)' = a^x \log a, a > 0$

Proof of 8: $(a^x)' = (e^{x \log a})' = a^x \log a$ (chain rule)

Product/quotient rules

If f, g are differentiable:

- $(f + g)' = f' + g'$
- $(fg)' = f'g + g'f$ [Leibniz rule]
- $\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2}$ where $g \neq 0$ near the point.

Example (revenue): $R(q) = p(q)q$, then $R'(q) = p'(q)q + p(q)$.

Chain rule

If $h = g \circ f$ and f, g are differentiable at the relevant points, then

$$h'(x) = g'(f(x)) f'(x).$$

Example (cost of production):

$$C(x) = 4 + \log(x + 1) + \sqrt{3x + 1} \quad \Rightarrow \quad C'(x) = \frac{1}{x + 1} + \frac{3}{2\sqrt{3x + 1}}.$$

Example (choice models): In discrete choice models, the logistic function $\sigma(x) = \frac{1}{1+e^{-x}}$ maps a score x (reflecting how attractive an option is) to the probability of choosing it. Its derivative $\sigma'(x) = \sigma(x)[1 - \sigma(x)]$ measures how sensitive the choice probability is to changes in x .

Differentiating $u(x)^{v(x)}$

Let $u(x) > 0$. For $h(x) = u(x)^{v(x)}$, taking logs gives:

$$\log h = v \log u \quad \Rightarrow \quad \frac{h'}{h} = v' \log u + v \frac{u'}{u}.$$

Hence

$$h'(x) = u(x)^{v(x)} \left(v'(x) \log u(x) + v(x) \frac{u'(x)}{u(x)} \right)$$

Econ example: A startup's valuation is $V(t) = V_0[1 + g(t)]^{T(t)}$, where $g(t)$ is the projected annual growth rate and $T(t)$ the number of years that growth lasts. Both $g(t)$ and $T(t)$ vary with t .

Some of the contributors (in form of *caganers*)



Newton



Leibniz



Euler



Lagrange

The inverse function theorem (1D version)

Theorem:

If f is strictly monotone, continuous, and differentiable at x with $f'(x) \neq 0$, then f^{-1} is differentiable at $y = f(x)$ and

$$(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{f'(f^{-1}(y))}.$$

Since $f^{-1}(f(x)) = x$, chain rule gives $(f^{-1})'(f(x))f'(x) = 1$.

Example: $f(x) = e^x \Rightarrow (\log)'(y) = 1/y$.

Higher-order derivatives: If f' is differentiable, its derivative is called the **second derivative** of f and denoted f'' . Similarly, we can define **higher-order derivatives**, and $f^{(n)}$ is called the n th derivative of f .

If $f^{(n)}$ is continuous, we say that f is n times **continuously differentiable** or **C^n** for short.

L'Hôpital's rule

Theorem: Let f, g be C^1 on (a, b) with $g' \neq 0$ on (a, b) . If

$$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} g(x) = 0 \quad \text{or} \quad \pm \infty,$$

then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} \quad (\text{if the RHS limit exists}).$$

Proof idea uses the Mean Value Theorem (slide 57).

Example:

$$\begin{aligned} \lim_{x \rightarrow 0} (1+x)^{1/x} &= \lim_{x \rightarrow 0} \exp\left(\frac{1}{x} \log(1+x)\right) \\ &= \exp\left(\lim_{x \rightarrow 0} \frac{\log(1+x)}{x}\right) \\ &= \exp\left(\lim_{x \rightarrow 0} \frac{1}{1+x}\right) = e. \end{aligned}$$

Optimization in 1D

Monotonicity via derivatives

If f is differentiable on (a, b) :

1. f increasing on $[a, b] \iff f'(x) \geq 0$ on (a, b) .
2. f decreasing on $[a, b] \iff f'(x) \leq 0$ on (a, b) .
3. f constant on $[a, b] \iff f'(x) = 0$ on (a, b) .
4. $f'(x) > 0$ on $(a, b) \Rightarrow f$ strictly increasing on $[a, b]$.
5. $f'(x) < 0$ on $(a, b) \Rightarrow f$ strictly decreasing on $[a, b]$.

Converse of 4-5 can fail at isolated points (e.g., x^3).

Example: $f(x) = \frac{x^3}{3} + 2x^2 + 3x + 1$; $f'(x) = (x + 1)(x + 3)$ determines monotonicity intervals and local extrema.

Optimal points: first-order conditions

Local optima

Let $f : D \rightarrow \mathbb{R}$. A point $x_0 \in D$ is a **local maximum** if $\exists (a, b) \subset D$ with $x_0 \in (a, b)$ and $f(x) \leq f(x_0)$ for all $x \in (a, b)$. Similarly for **local min** with $\geq$.

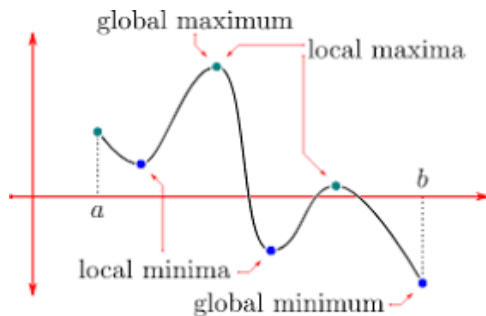
If f has a local optimum at interior x and is differentiable, then $f'(x) = 0$.

To see why this is true, stare at $f(x_0 + h) - f(x_0) \approx f'(x_0)h$.

This condition is **necessary**, not sufficient; see e.g. x^3 .

If x is stationary ($f'(x) = 0$) and f is strictly increasing to the left and strictly decreasing to the right, then x is a local max (reverse for min).

Optimal points



Example: Timing a sale (present value)

Suppose an assets market value is $V(t)$ at time t . At a constant annual interest rate R , the **present value** of selling at time t is

$$P(t) = V(t) e^{-Rt},$$

where e^{-Rt} discounts future cash flows.

Goal: Choose t to maximize $P(t)$.

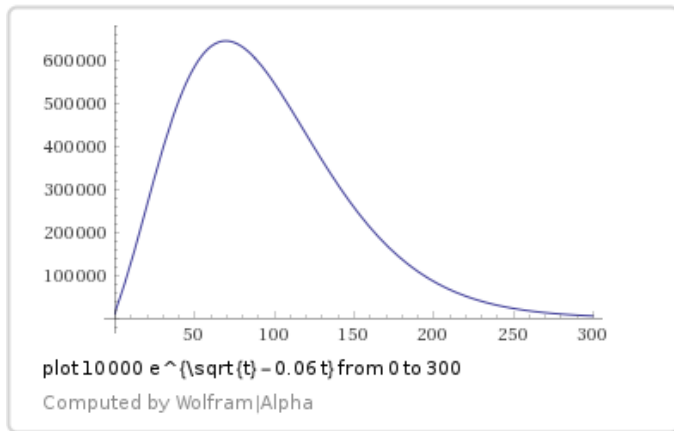
FOC (set derivative to zero):

$$P'(t) = V'(t)e^{-Rt} - R V(t)e^{-Rt} = 0 \quad \Rightarrow \quad \frac{V'(t)}{V(t)} = R.$$

Interpretation: Sell when the assets *instantaneous percentage growth rate* (matches) the *interest rate you could earn in the bank*.

Example: If $V(t) = 10000 e^{\sqrt{t}}$ and $R = 6\%$, then $P(t) = 10000 e^{\sqrt{t} - 0.06t}$, which is maximized near $t \approx 69.44$.

Plot of the present value $P(t)$



To certify a **global** optimum on an interval, compare values at boundaries and critical points. Here $P(0) = 10\,000$, $\lim_{t \rightarrow \infty} P(t) = 0$.

Second-order conditions

Test: If $f'(x_0) = 0$ and $f''(x_0) < 0$ (> 0), then x_0 is a strict local max (min).

$$\text{Indeed, } f(x_0 + h) \approx f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 = f(x_0) + \frac{1}{2}f''(x_0)h^2.$$

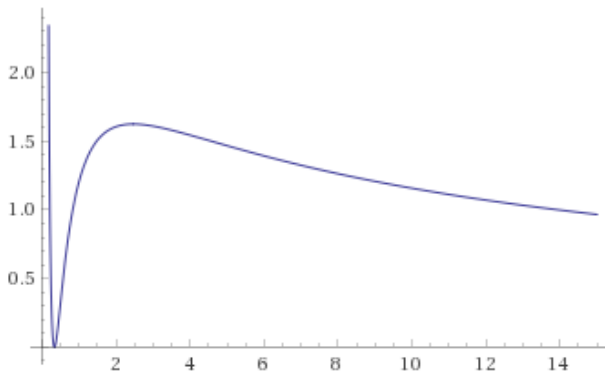
Example: For $f(x) = \frac{\log^2(3x)}{x}$ ($x > 0$):

$$f'(x) = \frac{(2 - \log 3x) \log 3x}{x^2}, \quad f''(x) = \frac{2(1 - 3 \log 3x + \log^2 3x)}{x^3}.$$

Critical points at $x_1 = \frac{1}{3}$ (min), $x_2 = \frac{e^2}{3}$ (max).

Question: What if $f''(x_0) = 0$?

Plot of $f(x) = \frac{\log^2 3x}{x}$, $x > 0$



plot $\ln^2(3x)/x$ from 0 to 15 | Computed by Wolfram|Alpha

$\lim_{x \rightarrow \infty} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = \infty$, $f(x_1) = 0$, $f(x_2) > 0 \Rightarrow x_1$ global min, no global max.

Convexity and concavity

If f is twice differentiable on (a, b) :

1. f convex on $[a, b] \Leftrightarrow f''(x) \geq 0$ for all $x \in (a, b)$.
2. f concave on $[a, b] \Leftrightarrow f''(x) \leq 0$ for all $x \in (a, b)$.
3. $f'' > 0$ (< 0) $\Rightarrow f$ strictly convex (concave).

Theorem: If f and g are convex (or concave), then $f \circ g$ is convex (or concave).

Example: $f(x) = e^{x^2}$ is strictly convex on $[0, \infty)$.

Economics:

- **Cost functions:** usually convex — producing extra units becomes progressively more expensive (e.g. overtime wages, machine wear).
- **Utility functions:** often concave — reflects diminishing marginal utility (the first slice of pizza gives more satisfaction than the tenth).
- **Machine learning:** convex losses (e.g. squared error, logistic loss + ℓ_2 regularization) ensure optimization problems are tractable and stable.

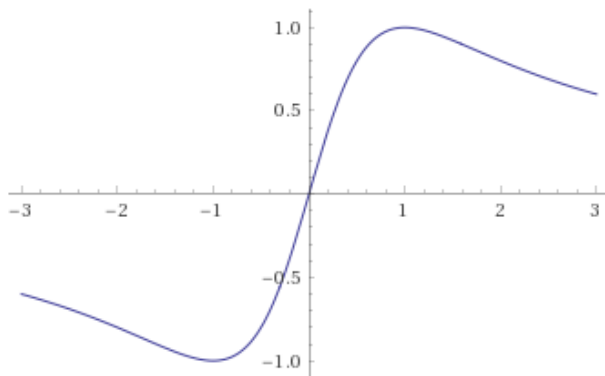
Example of changing curvature

Let $f(x) = \frac{2x}{x^2 + 1}$. Then

$$f'(x) = \frac{2(1 - x^2)}{(x^2 + 1)^2}, \quad f''(x) = \frac{4(x^3 - 3x)}{(x^2 + 1)^3}.$$

So f is strictly convex on $[-\sqrt{3}, 0] \cup [\sqrt{3}, \infty)$ and strictly concave on $(-\infty, -\sqrt{3}] \cup [0, \sqrt{3}]$.

Plot of $f(x) = \frac{2x}{x^2 + 1}$



plot $2x/(x^2+1)$ from -3 to 3 | Computed by Wolfram|Alpha

Inflexion points at $x \in \{-\sqrt{3}, 0, \sqrt{3}\}$. Global max at $x = 1$, min at $x = -1$, and $\lim_{|x| \rightarrow \infty} f(x) = 0$.

Inflexion points

If f is C^n on (a, b) , an **inflexion point** at x_0 occurs when curvature changes sign. A sufficient test:

$$f''(x_0) = \dots = f^{(n-1)}(x_0) = 0, \quad f^{(n)}(x_0) \neq 0, \quad n \text{ odd.}$$

Example: $f(x) = x^3$ has $f''(0) = 0$, $f^{(3)}(0) = 6 \neq 0 \Rightarrow$ inflexion at 0.

Taylor series expansion

Mean-value theorem

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad (\text{equivalently } f(b) = f(a) + f'(c)(b - a)).$$

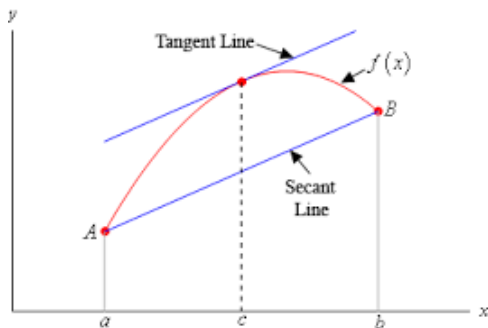
Proof idea: consider $h(x) = (f(b) - f(a))x - (b - a)f(x)$; since $h(a) = h(b)$, h has an interior extremum.

Interpretation: the instantaneous slope matches the secant slope at some point.

Geometry of the mean-value theorem

Recall: $\exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Taylor formula

Theorem (Taylor's formula of order n): Let f be $n + 1$ times differentiable on (a, b) , and let $x_0 \in (a, b)$ given. Then, for any $x \in (a, b)$, if $n = 1$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \text{error}.$$

If $n \geq 2$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \text{error}.$$

The error is small if x is close to x_0 , therefore, this theorem says that differentiable functions may be **locally approximated by a polynomial** called the **Taylor polynomial**.

Economics/data: local linear/quadratic approximations underlie elasticity (log-linearization), Delta method, and second-order price responses.

Taylor remainder and a handy expansion

Remainder (Lagrange form)

$$R_n^y(x) = \frac{f^{(n+1)}(y)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } y \text{ between } x_0 \text{ and } x.$$

If $|f^{(n+1)}| \leq M$ near x_0 , then $|R_n^y(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$.

Example (log around 1): With $x_0 = 1$,

$$\log x \approx (x - 1) - \frac{1}{2}(x - 1)^2, \quad R_2^y(x) = \frac{1}{3y^3}(x - 1)^3.$$

Finance quick check: For small r , $\log(1 + r) \approx r - \frac{r^2}{2} \Rightarrow \text{EAR} \approx \text{APR} - \frac{1}{2}\text{APR}^2$ (small correction).

Short comment about small-o notation

Writing $|R_n^y(x)| \leq \frac{M}{(n+1)!}|x-x_0|^{n+1}$ we conclude $\frac{R_n^y(x)}{|x-x_0|^n} \leq \frac{M}{(n+1)!}|x-x_0|$ and so

$$\lim_{x \rightarrow x_0} \frac{R_n^y(x)}{|x-x_0|^n} = 0.$$

This we write as $R_n^y(x) = o(|x-x_0|^n)$.

In particular: If f is twice differentiable:

$$f(x+h) = f(x) + f'(x)h + o(|h|).$$

The generalization of this last expression to multivariate functions will provide powerful insights into their optimization.